



Time Series Event Detection in R: Anomalies, Change Points, Motifs



Data Analytics Lab

dal.global@grupo.cefet-rj.br | <https://eic.cefet-rj.br/dal>



Harbinger: A Unified Framework

The Challenge

Fragmentation

Specialized tools scattered across different packages with no common interface or evaluation standard.

Real-World Demands

Practitioners need systematic comparison and combination of multiple detection methods within a single workflow.

Open-Source & Reproducible

Research and applications require a transparent, reusable ecosystem that promotes reproducibility across domains.

The Solution

Harbinger is a unified, open-source R package for anomaly detection, change point identification, and motif discovery — all in one coherent, reproducible ecosystem.



Stars 22 downloads 18K/month

Harbinger is a framework for event detection in time series. It provides an integrated environment for anomaly detection, change point detection, and motif discovery. Harbinger offers a broad range of methods and functions for plotting and evaluating detected events.

For anomaly detection, methods are based on: - Machine learning model deviation: Conv1D, ELM, MLP, LSTM, Random Regression Forest, and SVM - Classification models: Decision Tree, KNN, MLP, Naive Bayes, Random Forest, and SVM - Clustering: k-means and DTW - Statistical techniques: ARIMA, FBIAD, GARCH

For change point detection, Harbinger includes: - Linear regression, ARIMA, ETS, and GARCH-based approaches - Classic methods such as AMOC, ChowTest, Binary Segmentation (BinSeg), GFT, and PELT

For motif discovery, it provides: - Methods based on Hashing and Matrix Profile

Harbinger also supports **multivariate time series analysis** and **event evaluation** using both traditional and soft computing metrics.

The architecture of Harbinger is based on **Experiment Lines** and is built on top of the [DAL Toolbox](#). This design makes it easy to extend and integrate new methods into the framework.

50+

Detection Methods

Anomaly detection algorithms covering statistical, ML, and deep learning approaches

15+

Change Point Techniques

Methods for identifying structural breaks and regime changes in time series

10+

Motif Discovery Methods

Algorithms for finding recurring patterns and subsequences in temporal data

Modular Architecture

Design Principles

A strict interface grounded in workflow algebra — ensuring consistency and interoperability across all detection methods.

- Built on the DAL Toolbox, with design patterns inspired by the Scikit-learn API
- Familiar `fit()` and `detect()` functions provide an intuitive, low-friction interface

01

Normalize

Preprocessing and standardization

02

Filter

Noise reduction and signal enhancement

03

Fit

Model training and parameter estimation

04

Detect

Event identification and classification

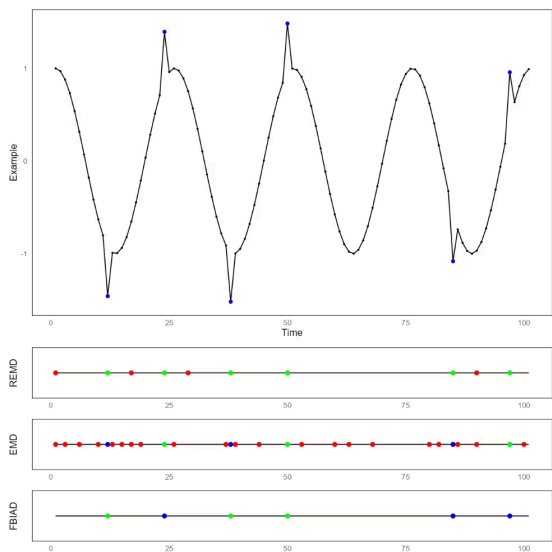
Detector Evaluation & Benchmarking

Rigorous evaluation quantifies **precision** and **recall** to reveal what a detector genuinely captures and what it misses. Harbinger enables systematic benchmarking across diverse datasets and detection scenarios:

- **Hard evaluation** — strict, exact match-based assessment
- **Soft evaluation** — tolerant scoring that accommodates temporal offsets

From Signal to Evidence

Evaluation transforms raw detector outputs into measurable, interpretable evidence of performance.



Temporal Tolerance

In time series, an event detected with a slight temporal offset can still be valuable — within an appropriate tolerance window.

Fair Comparison

Standardized metrics enable objective, apples-to-apples comparison of competing algorithms across benchmarks.

Precision & Recall

Quantify detection accuracy

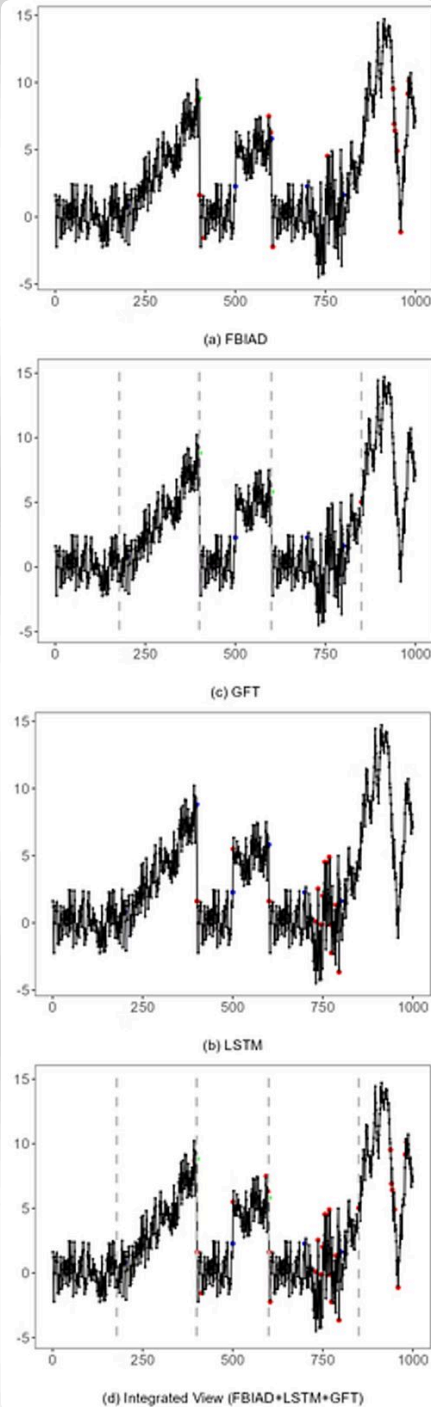
Temporal Alignment

Measure timing precision

Performance Metrics

Comprehensive end-to-end assessment

📄 `har_eval()` and `har_eval_soft()` ensure comparability, rigor, and consistency across all evaluation scenarios.



Combining Detectors: Ensemble Models

No single detector sees everything — ensemble methods combine complementary algorithms for more robust results.

- Each algorithm is sensitive to different patterns
- Combining signals suppresses false positives
- Consensus across methods increases confidence

Broader Coverage

Different detectors capture failures, patterns, and variations that any single method might overlook.

Consensus with Greater Confidence

Events confirmed by multiple algorithms carry stronger interpretive weight, making it easier to separate genuine signals from noise.

A Strategy Built for Critical Use Cases

Harbinger makes ensemble detection transparent and interpretable, supporting rigorous analysis in contexts where accuracy is non-negotiable.

```
model <- har_ensemble(
  hanr_fbiad(),
  hanr_arima(),
  hanr_emd())
model <- fit(model, dataset$serie)
detection <- detect(model, dataset$serie)
har_plot(model, dataset$serie,
  detection, dataset$event)
```

FBIAD

Frequency-based detection

ARIMA

Statistical modeling

EMD

Signal decomposition

Ensemble Aggregation

Combined decision output

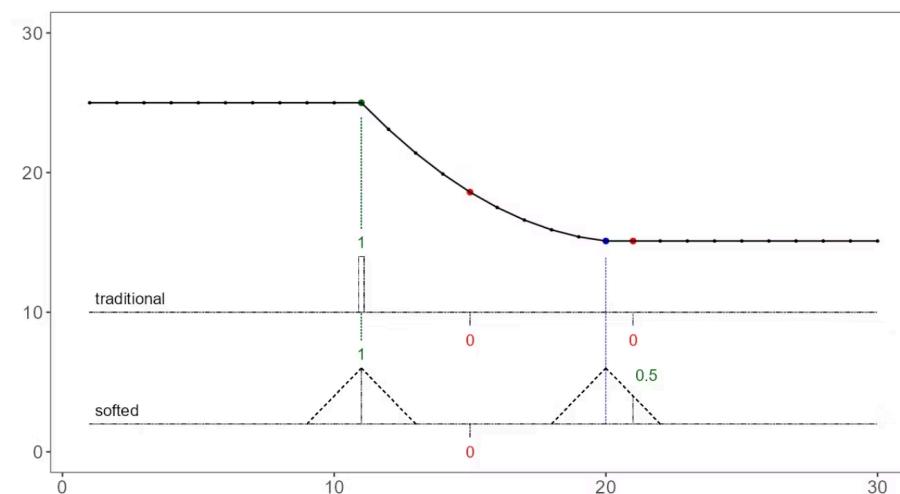
Fuzzifying Events: Beyond Binary Detection

SoftED Metrics

SoftED (Soft Event Detection) introduces fuzzy evaluation metrics that more faithfully capture how humans judge the quality of event detection.

Unlike traditional metrics, SoftED tolerates minor timing offsets while applying stronger penalties to fragmented detections.

Fuzzy membership functions deliver a graduated scale of quality, enabling far more nuanced assessment than a simple pass-or-fail verdict.



Temporal Tolerance

Accepts small timing misalignments without penalizing

Fragmentation Penalty

Penalizes broken or split detections

Graduated Scoring

Delivers nuanced, continuous quality assessment

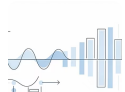
A Taxonomy of Time Series Event Detection

Time series event detection spans a rich diversity of methods — organized here by the types of events they target and their underlying algorithmic approaches.



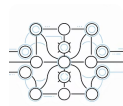
Statistical Methods

Leverage probabilistic models to characterize normal behavior and flag deviations



Signal Processing Techniques

Use frequency-domain analysis and decomposition to isolate distinctive event signatures



Machine Learning Approaches

Apply supervised or unsupervised learning to discover complex patterns in data



Hybrid Methods

Combine multiple paradigms to harness complementary strengths for greater accuracy

Acknowledgements



Data Analytics Lab

dal.global@grupo.cefet-rj.br | <https://eic.cefet-rj.br/dal>

